

QUESTION-TO-SLIDE MAPPING

SECTION B - LECTURE 02

Global Climatic Factors

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	2.0 - Global Climatic Factors (30 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Mapping rule	Map the 30-slide deck to the historical question bank by mechanism: solar radiation, Earth-Sun geometry, radiation at the surface, Earth thermal balance, global wind circulation, Coriolis effect, ITCZ/annual shift and topography. A question is marked Partial where the slide names or illustrates the issue but does not supply the full explanation required by the exam.

RESULT

Lecture 02 is one of the highest-return Section B decks. It directly supports the repeated global-climate families on Earth thermal balance, global wind patterns, ITCZ/annual wind shift, Earth-Sun geometry and radiation at the Earth surface. The main limitation is topography: Slide 27 introduces the interaction but does not fully develop all topographic modifiers asked in the question bank.

Prepared from the uploaded lecture deck and the supplied 2016-2024 regular final question bank.

1. Executive Summary

The lecture builds global climate from first principles: climate/weather, latitude-altitude and land-water effects; solar radiation quality and quantity; axial tilt and seasonal solar distribution; surface radiation and thermal balance; global pressure/wind circulation; thermal and Coriolis forces; ITCZ/doldrums; annual migration of wind belts; and a short topography note. Historically, this lecture connects to questions in nearly every supplied final paper and is especially diagram-heavy.

30	22	17	3	2	281
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Highest-priority slide blocks

Slides	Lecture block	Historical signals	Exam value
6-13	11 global factors; solar radiation quality/quantity; tilt; surface radiation; Earth thermal balance	2016 Q1(b), Q2(b); 2017 Q1(b); 2018 Q5(a); 2020 Q6(a); 2021 Q3(c)(i), Q4(c); 2022 Q2(b); 2023 Q2(a); 2024 Q1(a), Q4(c)	Very high. Core theory + diagrams + percentage model.
14-23, 28-29	Global wind cells, pressure belts, thermal force, Coriolis force, trades, westerlies, polar easterlies	2016 Q3(a); 2021 Q1(c); 2022 Q2(a,c); 2024 Q1(c)	Very high. Must reproduce a labelled 0-30-60-90 circulation sketch.
24-26	ITCZ, doldrums and annual north-south wind shift	2016 Q3(b); 2021 Q1(b); 2023 Q2(b); 2024 Q1(b)	Very high. Repeated conceptual family with seasonal sketch.
27	Influence of topography / land-water heating interaction	2021 Q1(a); 2023 Q4(a); 2024 related	Useful but incomplete for the broader topography questions.
2-5	Climate/weather, latitude-altitude, tropical belt, air mass	Foundation / later tropical-climate support	Low-medium as standalone final-question source; useful for introductions.

Priority note Safest order: 12-13 (thermal balance) -> 14-23 & 28-29 (global winds) -> 24-26 (ITCZ/annual shift) -> 9-11 (Earth-Sun and surface radiation) -> 8 (solar quantity). Practise the diagrams, not only the text.

2. Detailed Year-wise Question Mapping

The original printed section label may change from year to year. Mapping below follows subject matter and the lecture content. Slide numbering uses the PDF page count, including the title slide.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Q1(b)	25	1	Analyse five major factors of global climate variation in different geographic locations.	Exact	6-29	Slide 6 lists eleven global climatic factors and the following slides explain the major mechanisms in enough depth to select and analyse five.
2016	Q2(b)	15	1	How does the Earth keep thermal balance by equating incoming and outgoing radiation?	Exact	12-13	The lecture gives the incoming and outgoing percentage model and the three surface heat-release paths.
2016	Q3(a)	20	1	Evaluate the forces interacting to generate global wind patterns at different latitudes in both hemispheres.	Exact	14-23, 28-29	Pressure belts, circulation cells, thermal force, Coriolis effect and the main surface wind belts are directly shown.
2016	Q3(b)	15	1	Assess the impacts of annual wind shift from north to south of the equator.	Exact	24-26	ITCZ migration, north/south shift and seasonal changes in temperature, wind direction and rainfall are directly stated.
2017	Q1(b)	20	2	Evaluate tilt of the Earth axis from the viewpoint of global climate variation.	Direct	9-10	Axial tilt, solstice/equinox movement and long-cycle orbital variations are shown; the student must connect them to global climatic consequences.
2018	Q5(a)	15	3	In how many ways does the Earth surface release heat to the atmosphere? Discuss with illustrations.	Exact	12-13	Outgoing heat is shown as long-wave radiation, evaporation and convection with the lecture percentages.
2020	Q6(a)	15	5	How does the Earth keep thermal balance by equating incoming and outgoing radiation? Show with diagram.	Exact	12-13	Direct repetition of the lecture thermal-balance diagrams.
2021	Q1(a)	10	6	Assess the influence of topography from the perspective of global climate variations.	Partial	27	Slide 27 introduces global-regional interaction and land-water differential heating, but not the full set of slope, altitude, orographic and valley effects.
2021	Q1(b)	10	6	Describe the relationship between ITCZ and annual wind shifts with appropriate figures.	Exact	24-26	The ITCZ definition and seasonal north-south movement are the focus of these slides.
2021	Q1(c)	15	6	Explain global wind patterns at different latitudes in both hemispheres with neat sketches.	Exact	14-23, 28-29	The deck provides global belts, Hadley/Ferrel/Polar cells and surface wind directions.
2021	Q3(c)(i)	5	6	Write short note on solar radiation quantity.	Exact	8	Solar constant, variation, orbit period and aphelion/perihelion distances are directly given.

2. Detailed Year-wise Question Mapping - continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2021	Q4(c)	12	6	How does the Earth keep thermal balance by equating incoming and outgoing radiation?	Exact	12-13	Direct thermal-balance match.
2022	Q2(a)	10	8	Describe the relationship between thermal force and Coriolis force with appropriate figures.	Exact	21-23	The two forces are explicitly defined and linked to trade winds.
2022	Q2(b)	12	8	Evaluate Earth-Sun relationship from the viewpoint of global climate variation.	Direct	8-10	Solar distance/orbit and axial tilt are present; full climatic evaluation requires synthesis across the three slides.
2022	Q2(c)	13	8	Define trade wind. Write down characteristics of mid-latitude westerlies and polar winds.	Exact	14-23, 28-29	The global circulation sequence and Coriolis logic directly support the three wind families.
2023	Q2(a)	10	10	Evaluate radiation of the Earth surface at global scale.	Exact	11-13	Angle of incidence, atmospheric depletion, sunshine duration and thermal-balance paths are all present.
2023	Q2(b)	10	10	Define doldrums. Explain location and formation.	Exact	24	The lecture defines ITCZ and explicitly identifies the calm/irregular breeze zone as the doldrums.
2023	Q4(a)	5	10	Write down topographical factors responsible for global climate variation.	Partial	27	The slide only gives a brief continental interaction/land-water heating note, not a complete topographic-factor list.
2024	Q1(a)	8	13	Explain how tilt of the Earth axis affects distribution of solar radiation and causes seasonal changes.	Exact	9-10	Axial tilt, Tropics and equinox/solstice movement are directly illustrated.
2024	Q1(b)	12	13	Why is convection at the ITCZ generally calm, and why does it shift more toward the Northern Hemisphere?	Direct	24-26	Calm/doldrums and the greater northward shift are shown; the lecture states the asymmetry but does not fully explain the land-dominance reason.
2024	Q1(c)	15	13	Explain global wind patterns at different latitudes in both hemispheres with neat sketches.	Exact	14-23, 28-29	Direct repeated global-wind question.
2024	Q4(c)	9	13	Evaluate radiation of the Earth surface from the viewpoint of global climate variation.	Exact	11-13	The surface-radiation controls and thermal-balance model directly answer the question.

Coverage key

Exact = answer content is essentially present as asked. Direct = most content is present, but the student must apply/combine it. Partial = only one component is present; another lecture/source is required.

3. Repeated Question Patterns

Lecture 02 is not a collection of isolated short notes; it is a connected mechanism chain. Solar inequality creates pressure differences, pressure differences drive circulation, Earth rotation deflects it, and the seasonal solar maximum moves the ITCZ and wind belts. The exam repeatedly samples different points in this same chain.

P	Normalized repeated family	Years seen	Slides	Why it matters
S	Earth surface radiation + Earth thermal balance / heat release	2016, 2018, 2020, 2021, 2023, 2024	11-13	Most stable lecture-specific family; diagram + percentages.
S	Global wind pattern: thermal force, Coriolis, trades, westerlies, polar winds	2016, 2021, 2022, 2024	14-23, 28-29	Long-answer family; sketch carries marks.
S	ITCZ / doldrums / annual wind shift	2016, 2021, 2023, 2024	24-26	Repeated seasonal-circulation family.
A	Earth-Sun relationship / axial tilt / seasons	2017, 2022, 2024	8-10	High-return mechanism + seasonal diagram.
A	Solar radiation quantity	2021 + support in other years	8	Easy short note and useful opening for radiation answers.
B	Topography influence	2021, 2023 (+2024 related)	27	Exam asks broader detail than this deck provides.

4. Quick Recognition Rules

If the question says...	Go to slides	Recognition rule
"thermal balance / incoming-outgoing / Earth releases heat"	12-13	Use the 50% ground receipt model; outgoing paths = long-wave + evaporation + convection.
"global wind / thermal force / Coriolis / trade-wind / westerlies / polar"	14-23, 28-29	Start with pressure belts 0-30-60-90, then circulation + Coriolis deflection.
"ITCZ / doldrums / annual wind shift"	24-26	Trade winds converge -> air rises -> calm variable surface wind; seasonal heating shifts the belt N-S.
"Earth-Sun relationship / tilt / seasons"	8-10	Orbit/distance + 66.5° to orbital plane (23.5° from normal) + solstice/equinox movement.
"radiation at Earth surface"	11-13	Angle of incidence + atmospheric depletion + duration of sunshine, then surface energy balance.
"topography"	27 + Lecture 05/reference	Do not stop at Slide 27; add altitude, slope/aspect, orographic rain, wind channeling and valley effects from the proper source.

5. Slide-to-Question Reverse Index

Use this page while revising the lecture: start from a slide block, then immediately practice the linked past questions.

Slides	What the slides teach	Past-question links	Study action
2-5	Climate/weather; latitude-altitude; tropical belt; land-water; air mass	Foundation only	Read once; use for introductions, not as main revision block.
6-8	Eleven global factors + solar radiation quality/quantity	2016 Q1(b); 2021 Q3(c)(i)	Memorize the 11-factor list and solar constant data.
9-10	Earth axis tilt, seasons, eccentricity/obliquity/precession	2017 Q1(b); 2022 Q2(b); 2024 Q1(a)	Practise one clean solstice/equinox sketch.
11	Radiation at Earth surface: cosine law, depletion, sunshine duration	2023 Q2(a); 2024 Q4(c)	Use as mechanism page before thermal balance.
12-13	Incoming/outgoing thermal balance	2016 Q2(b); 2018 Q5(a); 2020 Q6(a); 2021 Q4(c)	Memorize percentages and redraw both diagrams.
14-23, 28-29	Global wind cells, pressure belts, thermal + Coriolis forces	2016 Q3(a); 2021 Q1(c); 2022 Q2(a,c); 2024 Q1(c)	Practise 0-30-60-90 labelled circulation from memory.
24-26	ITCZ, doldrums and annual shift	2016 Q3(b); 2021 Q1(b); 2023 Q2(b); 2024 Q1(b)	Draw July and January ITCZ positions.
27	Topography / differential heating note	2021 Q1(a); 2023 Q4(a) partial	Treat as a prompt to open Lecture 05/reference for full detail.

6. Coverage Gaps & Diagnostic Exclusions

The deck is exceptionally complete for global circulation, but two exam demands exceed what is printed: broad topographic analysis and the causal explanation for why the ITCZ often migrates farther north than south.

Gap	Affected questions	Action
Full topographic modifiers are not developed	2021 Q1(a); 2023 Q4(a)	Use Site Climate / reference material for altitude, slope/aspect, orographic rain, valleys and wind channeling.
Why northward ITCZ shift is stronger is not fully explained	2024 Q1(b)	Slides 25-26 show the asymmetry; use the standard land-water heating explanation only if it is in your class note/reference and label it as supporting context.
Tropical climate classification is only introduced, not classified	2018/2022/2023 tropical-climate questions	Use Lecture 04; Slides 4-5 are background only.
Detailed site/local climate deviation is outside this lecture	2024 Q3 and site-climate family	Use Lecture 05, not the global topography slide alone.

Mapping disciplineCourse-specific slide values are preserved exactly for exam consistency: Solar Constant = 1395 W/m²; surface-balance model = 23% diffuse + 27% direct to ground, with 20% long-wave + 20% evaporation + 10% convection returned from the ground. These are reported as lecture values, not silently replaced.

7. Recommended Study Plan

Study by mechanism rather than page order. If you can reproduce three diagrams - Earth-Sun seasons, Earth thermal balance, and global wind belts/ITCZ - you cover most of the repeated marks from this lecture.

Step	Study block	Slides	Practice immediately
1	Earth thermal balance + surface radiation	11-13	2016 Q2(b), 2018 Q5(a), 2020 Q6(a), 2023 Q2(a).
2	Global wind pattern + forces	14-23, 28-29	2016 Q3(a), 2021 Q1(c), 2022 Q2(a,c), 2024 Q1(c).
3	ITCZ / doldrums / annual shift	24-26	2016 Q3(b), 2021 Q1(b), 2023 Q2(b), 2024 Q1(b).
4	Earth-Sun geometry / tilt	8-10	2017 Q1(b), 2022 Q2(b), 2024 Q1(a).
5	Global-factor master list + topography gap	6, 27	2016 Q1(b), then supplement 2021/2023 topography from Lecture 05.

Exam-ready minimum

Priority	What to reproduce	Exam technique
A	Global wind belts sketch	Label 0°, 30°, 60°, 90°; pressure belts; trades; westerlies; polar easterlies; ITCZ.
A	Thermal-balance diagrams	Write incoming/outgoing percentages beside arrows; add one-line interpretation.
A	ITCZ seasonal sketch	Show July northward and January southward location plus seasonal wind/rain consequence.
A	Earth-Sun seasonal sketch	State 66.5° to orbital plane = 23.5° from normal; mark solstices/equinoxes.

Bottom line

Lecture 02 should be one of the first Section B decks you finish. Its repeated questions are highly diagrammatic and structurally predictable: radiation -> balance, unequal heating -> pressure/wind, Coriolis -> prevailing belts, seasonal heating -> ITCZ migration.